MILITARY SPECIFICATION SHEET

ELECTRON TUBE, PHOTOTUBE

TYPE 6363

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The complete requirements for procuring the electron tube described herein shall consist of this document and the latest issue of Specification MIL-E-1.

DESCRIPTION: Multiplier, 10 stage, head-on, S11 spectral response

DIMENSIONS AND PIN CONNECTIONS: See figure 1

ABSOLUTE RATINGS:

Parameter:	epp	eb-edy 10	edy 1-ek	edy 1-esd	Light source	Light
Unit:	v	v	---	---	---	lm
Maximum:	1,800 Note 1	200 Note 2	400 Note 3	Note 4	---	---
TEST CONDITIONS:	1,260	105	210	0	LS1	9.05×10^{-7}

ABSOLUTE RATINGS:

Parameter:	ib	Ib	Pp	TA	Cathode area	Alt
Unit:	mAdc	mAdc	W	°C	Lighted sq in.	ft
Maximum:	25	5	2.5	75	---	10,000
TEST CONDITIONS:	---	---	---	---	4.9	---

GENERAL:

Qualification - Required

© denotes changes

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METHOD	REQUIREMENT OR TEST	NOTES	CONDITIONS	SYMBOL	LIMITS		UNIT
					MINIMUM	MAXIMUM	
	Qualification inspection						
1331	Direct-interelectrode capacitance	5	Anode to dynode 10	C	0.5	3.0	pF
1331	Direct-interelectrode capacitance	-	Anode to all other elements	C	2.0	5.0	pF
5303	Spectral-response identification	-	S11	---	---	---	---
Ⓒ 1216	Base material insulating quality	-		---	---	---	---
	Quality conformance inspection, part 1	13					
---	Cathode current	6, 7	Light = 0.021 lm	Ik	0.84	---	μAdc
5301	Anode current	7, 8		Ib	3.64	---	μAdc
5301	Dark current (interelectrode)	8, 9		ΔIb	---	0.05	μAdc
5301	Dark current (anode)	8		ΔIb	---	0.05	μAdc
	Quality conformance inspection, part 2						
Ⓒ 1111	Base pin solder depth	10		---	---	---	---
Ⓒ 1101	Secureness of base, cap, or insert	-		---	---	---	---
Ⓒ 1105	Permanence of marking	-		---	---	---	---
	Quality conformance inspection, part 3						
---	Life test	11	Group D; t = 500 hours; Epp = 1,250 Vdc	---	---	---	---
---	Life-test end point:						
5301	Anode current	12		Ib	6	---	μAdc

NOTES:

1. Supply voltage (dc or peak ac). Overall anode to cathode.
2. Supply voltage (dc or peak ac). Anode to tenth dynode.
3. Supply voltage (dc or peak ac). First dynode to cathode.
4. The potential on the shield (focusing electrode) shall lie between the cathode and first dynode potentials.
5. All other elements shall be grounded.
6. Two hundred and ten volts shall be applied between cathode and all other electrodes. The pins shall be tied together.
7. The light during test shall cover the whole cathode area uniformly.
8. Tubes shall be tested under the following conditions: edy 1 - ek = 210 vdc; voltage on the multiplier structure = 105 vdc per stage; eb - edy 10 = 105 vdc; esd = edy 1.
9. The dark current to any one of the following electrodes shall conform to the requirements of this specification sheet: cathode, and dynode 1 through dynode 10.
10. Immersion in water shall be omitted and tubes shall be tested dry. Torque test shall be applied with 40.0 in.-lb minimum.
11. Initially, adjust luminous flux at cathode to obtain Ib = 8 μAdc. Color temperature shall be greater than 2,000°K.

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NOTES: -Continued

12. Anode current shall be measured under the same conditions as life test.

13. All quality conformance inspection, part 1, tests to be performed at the conclusion of the holding period.

Custodians:
Army - EL
Navy - EC
Air Force - 85

Review activities:
Air Force - 11, 17, 80
DSA - ES

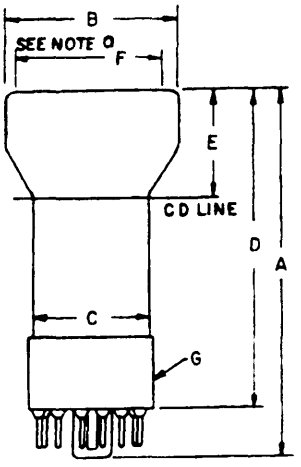
User activities:
Army - WC
Navy - AS, OS, MC, CG, SH
Air Force -

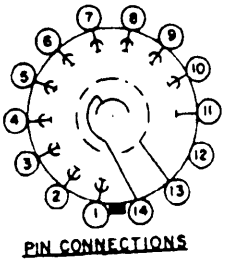
Preparing activity:
Navy - EC

Agent:
DSA - ES

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Ltr	Dimensions in inches with metric equivalents (mm) in parentheses	
	Minimum	Maximum
Qualification inspection		
G	Base: B14-38	
Quality conformance inspection, part 2		
A	5.940 (150.88)	6.310 (160.27)
B	2.906 (73.81) Dia.	3.094 (78.59) Dia.
C	1.938 (49.23) Dia.	2.063 (52.40) Dia.
D	5.190 (131.83)	5.560 (141.22)
E	1.560 (39.62)	1.940 (49.28)
F	2.500 (63.50)	---

PIN NO.	ELEMENT
1	Dynode No. 1
2	Dynode No. 2
3	Dynode No. 3
4	Dynode No. 4
5	Dynode No. 5
6	Dynode No. 6
7	Dynode No. 7
8	Dynode No. 8
9	Dynode No. 9
10	Dynode No. 10
11	Anode
12	Internal connection
13	Shield
14	Cathode

NOTES:
a. Minimum photocathode diameter.
b. Direction of light is into end of bulb.

FIGURE 1. Outline drawing of electron tube type 6363.

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NAVAL ELECTRONIC SYSTEMS COMMAND
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